



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Llama3-Driven Fault Detection and Decision Engine for Smart Governance in Power Systems

An Interdisciplinary Project Report (CS367P)

Submitted by,

Samkit Samsukha

1RV23IS105

Oojam Chaudhary

1RV23IS079

Shreeya Suman

1RV23EE048

Shradha Nageshwar

1RV23EE046

Under the guidance of

Prof. Rekha B S

Assistant Professor

Dept. of ISE

RV College of Engineering®

**In partial fulfillment of the requirements for the degree of
Bachelor of Engineering in respective departments**

2025-26



RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)



CERTIFICATE

Certified that the interdisciplinary project (CS367P) work titled ***Llama3-Driven Fault Detection and Decision Engine for Smart Governance in Power Systems*** is carried out by _____ of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

Prof. Rekha B S
Guide

Dr. Shanta Rangaswamy
Head of the Department

Dr. M.V. Renukadevi
Dean Academics

Dr. K. N. Subramanya
Principal

External Viva

Name of Examiners

Signature with Date

- 1.
- 2.

DECLARATION

We, **Samkit Samsukha, Oojam Chaudhary, Shreeya Suman, Shradha Nageshwar** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**Llama3-Driven Fault Detection and Decision Engine for Smart Governance in Power Systems**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

1. Samkit Samsukha(1RV23IS105)
2. Oojam Chaudhary(1RV23IS079)
3. Shreeya Suman(1RV23EE048)
4. Shradha Nageshwar(1RV23EE046)

ACKNOWLEDGEMENTS

We are indebted to our guide, **Prof. Rekha B S**, Assistant Professor, Department of ISE, RVCE for the wholehearted support, suggestions and invaluable advice throughout our project work and also helped in the preparation of this thesis.

We also express our gratitude to our panel member **Dr. Prapulla S B**, Associate Professor, Department of CSE, RVCE for the valuable comments and suggestions during the phase evaluations.

Our sincere thanks to the project coordinator **Dr. Prapulla S B**, Associate Professor, Department of CSE for timely instructions and support in coordinating the project.

Our gratitude to **Prof. Narashimaraja P**, Department of ECE, RVCE for the organized latex template which made report writing easy and interesting.

Our sincere thanks to **Dr. Shanta Rangaswamy**, Professor and Head, Department of Information Science and Engineering, RVCE for the support and encouragement.

We are deeply grateful to **Dr. M.V. Renukadevi**, Professor and Dean Academics, RVCE, for the continuous support in the successful execution of this Interdisciplinary project.

We express sincere gratitude to our beloved Professor and Vice Principal, **Dr. Geetha K S**, RVCE and Principal, **Dr. K. N. Subramanya**, RVCE for the appreciation towards this project work.

Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

Transmission lines are critical components of modern power systems, responsible for delivering electricity from generation units to consumers. However, they are highly susceptible to faults caused by environmental disturbances, equipment failures, and load variations. These faults can lead to system instability, equipment damage, and large-scale power outages if not detected and handled promptly. Traditional fault detection methods, primarily based on relay systems, often lack adaptability and fail to provide deeper insights into system-wide impacts. With the increasing complexity of smart grids, there is a growing need for intelligent, automated, and real-time monitoring solutions.

This report presents an advanced intelligent fault detection and predictive governance system that integrates Embedded Machine Learning, Decision Intelligence, and Generative AI. The system utilizes real-time voltage and current sensor data to detect and classify faults using a lightweight Decision Tree model deployed on embedded hardware. Beyond detection, a Decision Intelligence Engine analyzes cascading effects and evaluates system-wide impact, enabling proactive risk assessment. A Llama 3-based Generative AI layer is incorporated to perform root cause analysis, generate natural language explanations, and suggest mitigation strategies. Additionally, a Smart Governance Layer automates alerts, escalation workflows, and policy-driven responses, ensuring efficient and structured fault management.

The proposed system demonstrates high fault classification accuracy with low latency suitable for real-time applications. The integration of decision intelligence and AI-driven interpretation significantly enhances system usability and operational efficiency compared to traditional approaches. Furthermore, the system introduces a predictive governance paradigm by mapping fault events to potential operational and economic impacts, enabling better decision-making. The overall architecture provides a scalable and practical solution for intelligent power system monitoring, with potential applications in smart grid management and industrial energy systems.

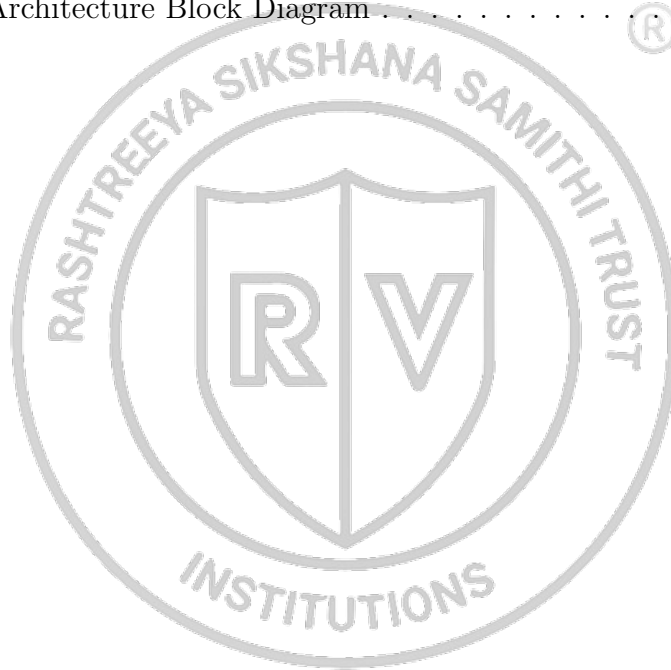
CONTENTS

Abstract	i
List of Figures	iv
List of Tables	v
1 A Hybrid Approach for Dysarthric Speech Recognition and Correction	1
1.1 Introduction	2
1.2 Literature Review	3
1.3 Motivation	3
1.4 Problem Statement	4
1.5 Objectives	4
1.6 Brief Methodology of the Project	4
1.7 Assumptions and Constraints	4
1.8 Organization of the Report	5
2 Fundamentals of Dysarthric Speech Recognition Systems	6
2.1 Speech and Dysarthria Fundamentals	7
2.2 Automatic Speech Recognition (ASR)	7
2.3 Speech Feature Extraction	8
2.4 Data Augmentation and Synthetic Speech	9
2.5 Deep Learning Models and Fine-Tuning	9
2.6 Evaluation Metrics	10
2.7 Programming Environment and Tools	10
2.8 Summary	11
3 System Design and Architecture for Dysarthric Speech Recognition	12
3.1 Specifications for the Design	13
3.2 Pre-analysis and Model Selection	13
3.3 Design Methodology	14
3.4 Experimental Techniques	15
3.5 Summary	16

4	Implementation of Speech Recognition Enhancement using Whisper	17
4.1	Contents of this chapter	18
4.2	System Overview	18
4.3	Baseline Evaluation	18
4.4	Data Augmentation	19
4.5	Synthetic Data Generation	19
4.6	Model Fine-Tuning	20
4.7	LLM-based Post Processing	20
4.8	Top Level Design	20
4.9	Performance Evaluation	21
4.10	Summary	21
5	Results & Discussions	23
5.1	Contents of this chapter	24
5.2	Simulation Results	24
5.3	Experimental Results	24
5.4	Performance Comparison	24
5.5	Comparison with Existing Models	25
5.6	Mathematical Formulation	25
5.7	Interpretation of Results	26
6	Conclusion and Future Scope	27
6.1	Conclusion	28
6.2	Future Scope	28
6.3	Learning Outcomes of the Project	29
A	Code	30
A.1	First Appendix	31

LIST OF FIGURES

1.1	ASR Pipeline Comparison: Normal vs. Dysarthric Speech Recognition . . .	2
2.1	Comparison of Speech Waveform and Spectrogram Representations . . .	8
2.2	Mel Spectrogram Example: Frequency-Time Representation of Speech . .	9
2.3	Whisper Encoder-Decoder Transformer Architecture for End-to-End ASR	10
3.1	System Pipeline Flow	15
3.2	Data Pipeline	15
4.1	System Architecture Block Diagram	21



LIST OF TABLES

5.1	WER comparison of baseline and fine-tuned model	25
5.2	WER results for different ASR architectures on TORGO and UASpeech .	25



Chapter 1

**An Intelligent Predictive
Governance System for Real-Time
Fault Detection in Power
Transmission Lines**

CHAPTER 1

AN INTELLIGENT PREDICTIVE GOVERNANCE SYSTEM FOR REAL-TIME FAULT DETECTION IN POWER TRANSMISSION LINES

1.1 Introduction

Electrical power transmission systems form the backbone of modern infrastructure, ensuring uninterrupted delivery of energy from generation units to end consumers. However, transmission lines are highly susceptible to faults caused by environmental disturbances, equipment failures, insulation breakdowns, and load fluctuations. Rapid and accurate fault detection is critical to prevent cascading failures, minimize downtime, and maintain grid stability.

Traditional protection systems rely on relay-based mechanisms that often lack adaptability under dynamic grid conditions. While recent advancements in machine learning have enabled real-time fault detection using sensor data, these systems remain limited to classification and lack higher-level intelligence for decision-making and system-wide impact analysis.

This project extends beyond conventional fault detection by introducing a multi-layered intelligent system integrating Embedded Machine Learning, Decision Intelligence, and Generative AI. The system not only detects faults in real time but also analyzes cascading effects, predicts operational risks, and provides actionable insights through natural language explanations.

By combining edge-based inference with AI-driven governance, the system transforms reactive monitoring into proactive and predictive grid management.

1.2 Literature Review

Recent research in transmission line fault detection has explored various machine learning approaches. Studies show that models such as Naive Bayes, Random Forest, and Decision Trees achieve high accuracy in classifying faults using voltage and current datasets. Decision Trees, in particular, are preferred for embedded systems due to their low computational complexity and fast inference.

Two-stage frameworks separating fault detection and classification have been proposed

to improve reliability. Ensemble learning techniques have demonstrated high accuracy in complex fault scenarios, though they often require significant computational resources, making them less suitable for real-time embedded deployment.

Other research has focused on fault localization using regression models, achieving high precision in estimating fault distance along transmission lines. However, most existing systems are limited to detection and classification, without addressing broader system-level implications.

Recent advancements in Artificial Intelligence have introduced Large Language Models (LLMs) for reasoning and decision support. These models enable contextual understanding, root cause analysis, and human-like explanations. However, their integration into power system monitoring remains largely unexplored.

Overall, existing work lacks a unified system that combines real-time detection, system-wide impact analysis, and intelligent decision-making.

1.3 Motivation

Modern power grids are becoming increasingly complex, with interconnected systems where a single fault can trigger cascading failures across multiple nodes. Traditional monitoring systems are reactive, addressing faults only after they occur, leading to delays, operational risks, and economic losses.

There is a growing need for systems that can not only detect faults but also understand their implications, predict potential disruptions, and assist operators in making informed decisions.

This project is motivated by the vision of transforming power grid monitoring into an intelligent, proactive, and self-assisting system. By integrating embedded machine learning with Generative AI and decision intelligence, the system aims to provide real-time insights, automate governance processes, and enhance operational efficiency.

1.4 Problem Statement

Existing transmission line monitoring systems are limited to fault detection and classification, lacking capabilities for system-wide impact analysis, predictive risk assessment, and intelligent decision support. This results in delayed responses, inefficient resource allocation, and increased risk of cascading failures in power grids.

1.5 Objectives

The objectives of the project are:

1. To develop a real-time fault detection system using embedded machine learning on voltage and current sensor data
2. To design a Decision Intelligence Engine that evaluates cascading effects and system dependencies
3. To integrate Llama 3-based Generative AI for fault interpretation, root cause analysis, and mitigation strategy generation
4. To implement a Smart Governance Layer for automated alerts, escalation workflows, and policy-driven responses
5. To build a web-based dashboard for real-time monitoring, analytics, and AI-driven insights

1.6 Brief Methodology of the Project

The proposed system follows a multi-layered intelligent architecture.

At the edge level, voltage and current sensors continuously capture real-time transmission line data. This data is processed by an embedded system where a Decision Tree model performs low-latency fault classification.

Once a fault is detected, the information is passed to a Decision Intelligence Engine, which analyzes system-wide dependencies, evaluates cascading effects, and assigns a risk score based on the severity and potential impact.

A Generative AI layer powered by Llama 3 interprets the fault context, performs root cause analysis, and generates mitigation strategies in natural language. This enables human operators to understand complex fault scenarios quickly and take informed actions.

The Smart Governance Layer automates alerts, escalation mechanisms, and policy-based responses, ensuring timely intervention and structured decision-making.

Finally, a web-based dashboard built using React and Flask provides real-time visualization, historical analytics, and AI-driven insights for operators.

This approach transforms the system from simple fault detection to an intelligent predictive governance framework.

1.7 Assumptions and Constraints

- Real-time sensor data is assumed to be accurate and properly calibrated
- Embedded systems have limited computational and memory resources
- Decision Tree models are chosen for efficiency, though they may have slightly lower accuracy than complex models
- LLM-based analysis depends on the quality of input data and contextual information
- Simulation of large-scale grid behavior is limited in a prototype environment

1.8 Organization of the Report

This report is organized as follows:

- Chapter 2 discusses fundamental concepts of power systems, fault detection, and machine learning techniques
- Chapter 3 presents the system architecture, including embedded ML, decision intelligence, and AI integration
- Chapter 4 details implementation, hardware setup, and software components
- Chapter 5 presents results, performance evaluation, and system analysis
- Chapter 6 concludes the work and outlines future enhancements in predictive governance and smart grid systems

Chapter 2
**Fundamentals of Intelligent Fault
Detection and Predictive
Governance Systems**

CHAPTER 2

FUNDAMENTALS OF INTELLIGENT FAULT DETECTION AND PREDICTIVE GOVERNANCE SYSTEMS

Modern power systems are rapidly evolving into complex, interconnected networks that require intelligent monitoring and decision-making capabilities. Traditional fault detection systems are no longer sufficient to handle dynamic grid conditions and cascading failures. Understanding the foundational concepts of electrical fault detection, embedded machine learning, decision intelligence, and generative AI is essential for designing advanced smart grid systems.

This chapter presents the core concepts required for the project, including transmission line fault fundamentals, machine learning models for real-time classification, system-wide decision intelligence, and AI-driven reasoning. It also introduces the tools and technologies used for implementation, providing a foundation for the system architecture discussed in the next chapter.

2.1 Transmission Line Fault Fundamentals

Transmission lines are critical components of the power system, responsible for delivering electricity over long distances. Faults in these lines can occur due to environmental conditions, insulation failure, equipment malfunction, or sudden load variations.

Common types of faults include:

- Single Line-to-Ground (L-G) faults
- Line-to-Line (L-L) faults
- Double Line-to-Ground (L-L-G) faults
- Three-phase faults

These faults result in abnormal voltage and current patterns, which can be detected using sensors. Rapid identification is essential to prevent equipment damage, reduce downtime, and maintain grid stability.

2.2 Real-Time Fault Detection using Embedded Systems

Modern fault detection systems use embedded hardware such as microcontrollers (e.g., ESP32 or Raspberry Pi) to process real-time sensor data. Voltage and current sensors continuously monitor line conditions, and analog signals are converted into digital data using ADC modules.

Embedded systems enable:

- Low-latency data processing
- Continuous real-time monitoring
- Deployment of lightweight machine learning models

This approach ensures faster response compared to centralized systems and reduces dependency on high-latency cloud processing.

2.3 Machine Learning for Fault Classification

Machine learning models are used to classify faults based on patterns in voltage and current data. Among various models, Decision Trees are widely used for embedded systems due to their simplicity, interpretability, and low computational requirements.

The classification process involves:

- Training on labeled voltage-current datasets
- Learning patterns associated with different fault types
- Performing real-time inference on incoming sensor data

Decision Trees provide fast and reliable predictions, making them suitable for deployment on resource-constrained devices.

2.4 Decision Intelligence in Power Systems

While fault detection identifies the type of fault, it does not provide insight into its broader impact. Decision Intelligence extends this capability by analyzing system-wide dependencies and predicting cascading effects.

Key aspects include:

- Modeling grid interdependencies
- Evaluating cascading failures across connected nodes
- Assigning risk scores based on severity and impact
- Supporting decision-making for fault mitigation

This layer transforms the system from reactive monitoring to proactive analysis and planning.

2.5 Generative AI for Fault Interpretation

Generative AI models, such as Llama 3, enable advanced reasoning capabilities in the system. These models process fault data and generate human-readable explanations, root cause analysis, and mitigation strategies.

Capabilities include:

- Natural language interpretation of faults
- Root cause identification based on historical and real-time data
- Step-by-step mitigation recommendations
- Interactive querying for operators

This allows operators to quickly understand complex fault scenarios and take informed actions.

2.6 Smart Governance and Automation

Smart Governance introduces automation into the system by enforcing policies and triggering actions based on predefined conditions.

Key features include:

- Automated alerts and notifications
- Escalation workflows based on fault severity
- Policy-driven responses for critical conditions
- Logging and reporting for compliance and analysis

This ensures structured and efficient handling of faults, reducing manual intervention and response time.

2.7 Evaluation Metrics

The performance of the system is evaluated using:

- Fault Detection Accuracy
- Classification Accuracy
- Latency (response time)
- System Reliability

These metrics help assess the effectiveness of real-time detection, decision-making, and system responsiveness.

2.8 Programming Environment and Tools

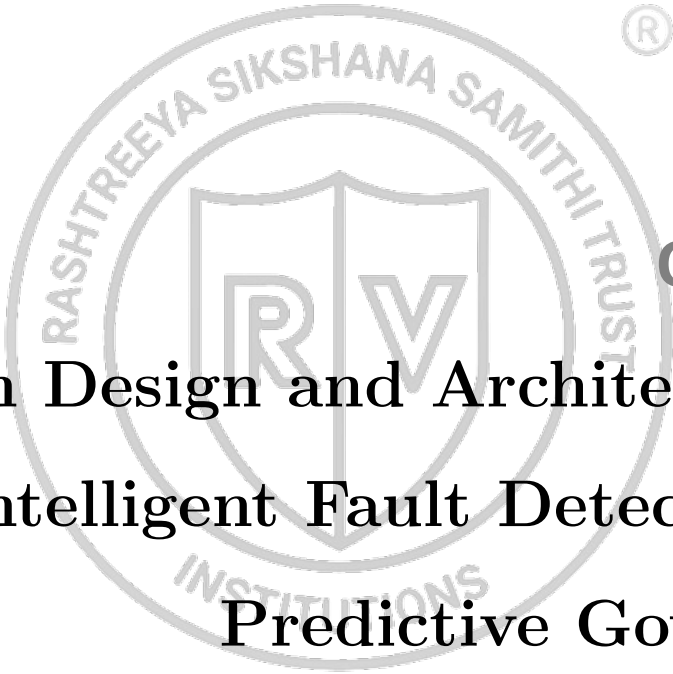
The implementation of the project uses:

- Python for machine learning and backend development
- Scikit-learn, NumPy, and Pandas for data processing and modeling
- Llama 3 and LangChain for AI-based reasoning and interaction
- Flask for backend APIs and data handling
- PostgreSQL for structured data storage
- React.js for building the monitoring dashboard
- Raspberry Pi OS / Embedded systems for real-time deployment

These tools provide a complete ecosystem for building an intelligent, real-time fault detection and decision system.

2.9 Summary

This chapter introduced the fundamental concepts required for intelligent fault detection and predictive governance systems, including transmission line fault behavior, embedded machine learning, decision intelligence, and generative AI integration. It also outlined the tools and technologies used for implementation. These foundations support the system architecture and design methodology presented in the next chapter.



Chapter 3

System Design and Architecture for Intelligent Fault Detection and Predictive Governance

CHAPTER 3

SYSTEM DESIGN AND ARCHITECTURE FOR INTELLIGENT FAULT DETECTION AND PREDICTIVE GOVERNANCE

Designing an intelligent fault detection system for power transmission lines requires careful consideration of real-time constraints, system scalability, and decision-making capabilities. The system must process high-frequency sensor data, accurately classify faults, and provide meaningful insights for operators. This chapter presents the design specifications, modeling choices, and architecture of the proposed multi-layered system. It also details the methodology and design considerations involved in integrating embedded machine learning, decision intelligence, and generative AI.

3.1 Specifications for the Design

The system is designed with the following specifications:

- Input: Real-time voltage and current sensor data from transmission lines
- Output: Fault classification, risk score, and AI-generated insights
- Model: Decision Tree for embedded real-time inference
- Processing: Edge-based inference using embedded systems (ESP32/Raspberry Pi)
- Intelligence: Decision Engine for cascading analysis and Llama 3 for interpretation
- Robustness: Ability to handle noisy sensor data and varying load conditions
- Evaluation: Performance measured using classification accuracy, latency, and system reliability

3.2 Pre-analysis and Model Selection

The design begins with evaluating traditional fault detection methods and machine learning models. While complex models such as Random Forest and Neural Networks offer high accuracy, they are not suitable for embedded deployment due to computational overhead.

Decision Trees are selected as the primary model due to:

- Low computational complexity
- Fast inference suitable for real-time applications
- High interpretability

To enhance system capabilities beyond classification, the following components are incorporated:

- Embedded ML for real-time fault detection
- Decision Intelligence Engine for cascading effect analysis
- Generative AI (Llama 3) for fault interpretation and mitigation
- Smart Governance Layer for automated alerts and policy enforcement

This results in a hybrid architecture combining edge intelligence with AI-driven reasoning.

3.3 Design Methodology

The proposed system follows a multi-stage intelligent pipeline:

1. **Data Acquisition:** Voltage and current sensors continuously capture real-time transmission line data
2. **Signal Processing:** Analog signals are converted into digital form using ADC modules and preprocessed for noise reduction
3. **Fault Classification:** A Decision Tree model deployed on embedded hardware classifies faults in real time
4. **Decision Intelligence:** The system analyzes grid dependencies, evaluates cascading effects, and assigns risk scores
5. **AI Interpretation:** Llama 3 processes fault data to generate root cause analysis and mitigation strategies
6. **Governance and Automation:** Alerts, escalation workflows, and policy-driven responses are triggered automatically

7. **Visualization:** A web-based dashboard displays real-time data, insights, and system status

This structured pipeline ensures both fast detection and intelligent decision-making.

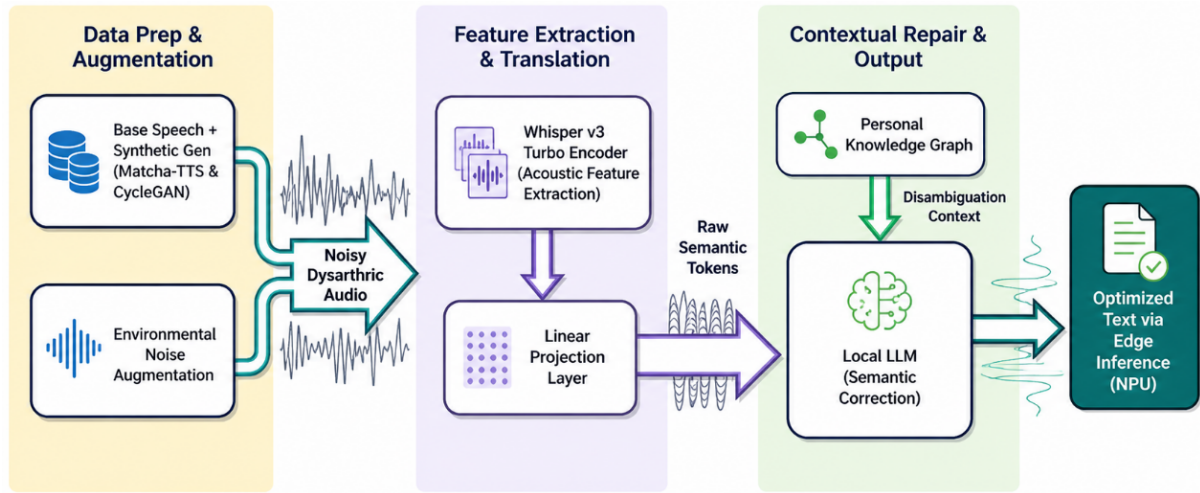


Figure 3.1: Intelligent Fault Detection and Decision Pipeline

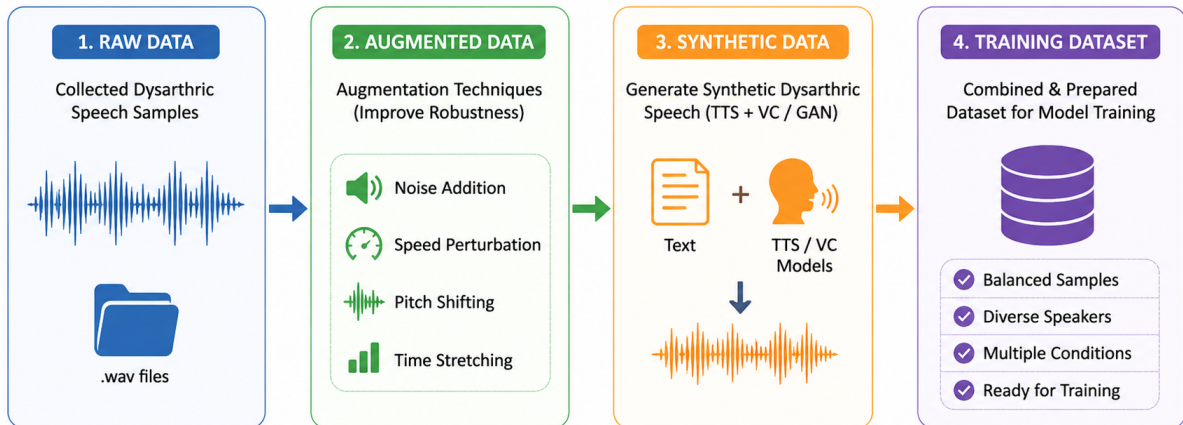


Figure 3.2: Sensor Data Acquisition and Processing Pipeline

3.4 Experimental Techniques

The system is developed and evaluated using:

- Real-time data acquisition from sensors under normal and fault conditions
- Simulation of faults using controlled switching and variable loads
- Training and testing of the Decision Tree model on labeled datasets

- Evaluation of system latency and real-time performance
- Validation of AI-generated insights and decision engine outputs
- Comparison between traditional detection and intelligent system outputs

3.5 Summary

This chapter presented the design and architecture of the proposed intelligent fault detection and predictive governance system. It outlined the specifications, model selection, and methodology used to build a multi-layered pipeline combining embedded machine learning, decision intelligence, and generative AI. These design principles guide the implementation and evaluation discussed in the next chapter.



Chapter 4
**Implementation of Intelligent Fault
Detection and Predictive
Governance System**

CHAPTER 4

IMPLEMENTATION OF INTELLIGENT FAULT DETECTION AND PREDICTIVE GOVERNANCE SYSTEM

The implementation integrates real-time data acquisition, embedded machine learning, decision intelligence, and generative AI into a unified pipeline for intelligent fault detection in transmission lines. The process begins with establishing baseline fault classification performance, followed by integration of real-time sensor data and embedded inference. Advanced decision analysis and AI-based interpretation layers are incorporated to enhance system intelligence and usability. Finally, a smart governance mechanism automates alerts and responses, enabling proactive system management. The chapter concludes with evaluation insights demonstrating improvements in detection speed, decision-making, and operational efficiency.

4.1 Contents of this chapter

This chapter elaborates the following:

1. Implementation details for the hardware/software pipeline used in the project
2. Top level design and workflow of the intelligent fault detection system

4.2 System Overview

The system focuses on real-time fault detection and intelligent decision-making in power transmission lines using a structured pipeline. The implementation is divided into five major components: baseline evaluation, real-time data acquisition, embedded fault classification, decision intelligence analysis, and AI-based interpretation with governance automation. Each component contributes to improving system reliability, responsiveness, and usability.

4.3 Baseline Evaluation

The initial step involves evaluating the performance of the trained Decision Tree model on labeled voltage-current datasets under simulated conditions.

The evaluation metrics used include classification accuracy and latency.

The observed results are:

- Fault Classification Accuracy: High accuracy for major fault types (above 95%)
- Detection Latency: Low response time suitable for real-time applications
- Stable performance under controlled test conditions

These results validate the suitability of Decision Trees for embedded real-time fault detection.

4.4 Real-Time Data Acquisition

To enable real-time monitoring, voltage and current data are continuously captured using hardware sensors.

Key components include:

- Voltage Sensors (ZMPT101B) for measuring line voltage
- Current Sensors (ACS712) for detecting current variations
- MCP3008 ADC for analog-to-digital conversion
- ESP32/Raspberry Pi for data processing

Sensor data is sampled continuously and transmitted to the embedded system for further processing and analysis.

4.5 Embedded Fault Classification

The Decision Tree model is deployed on the embedded system to perform real-time fault classification.

Key implementation details:

- Lightweight model optimized for embedded deployment
- Continuous inference on streaming sensor data
- Classification of multiple fault types based on voltage-current patterns

This ensures fast and reliable detection with minimal computational overhead.

4.6 Decision Intelligence Engine

Once a fault is detected, the system analyzes its broader impact using a Decision Intelligence Engine.

Key functionalities include:

- Evaluation of cascading effects across the grid
- Analysis of system dependencies and affected nodes
- Assignment of risk scores based on severity and impact

This component enables proactive decision-making rather than reactive fault handling.

4.7 LLM-based Interpretation and Insights

A Generative AI layer powered by Llama 3 is integrated to enhance system intelligence. The AI module performs:

- Root cause analysis of detected faults
- Natural language explanation of fault conditions
- Generation of mitigation strategies for operators

This improves system usability by translating technical outputs into actionable insights.

4.8 Smart Governance Layer

To automate system response, a governance layer is implemented.

The module includes:

- Automated alerts and notifications
- Escalation workflows based on risk levels
- Policy-driven actions for critical fault scenarios
- Logging and report generation for analysis and compliance

This ensures structured and efficient handling of faults in real time.